

scans, interpreting genomes or identifying cardiovascular risk, AI is proving most valuable in cases where data is abundant, workflows are structured and shortage of specialists is acute.

Policy think-tank NITI Aayog says AI has the potential to significantly accelerate India's economic growth by contributing to a three-fold increase in Gross Domestic Product (GDP) by 2035.

Kalyan Sivasailam, Co-founder and CEO of 5C Networks, a health-tech start-up that builds AI-powered diagnostic infrastructure, primarily for radiology, goes deeper into the topic.

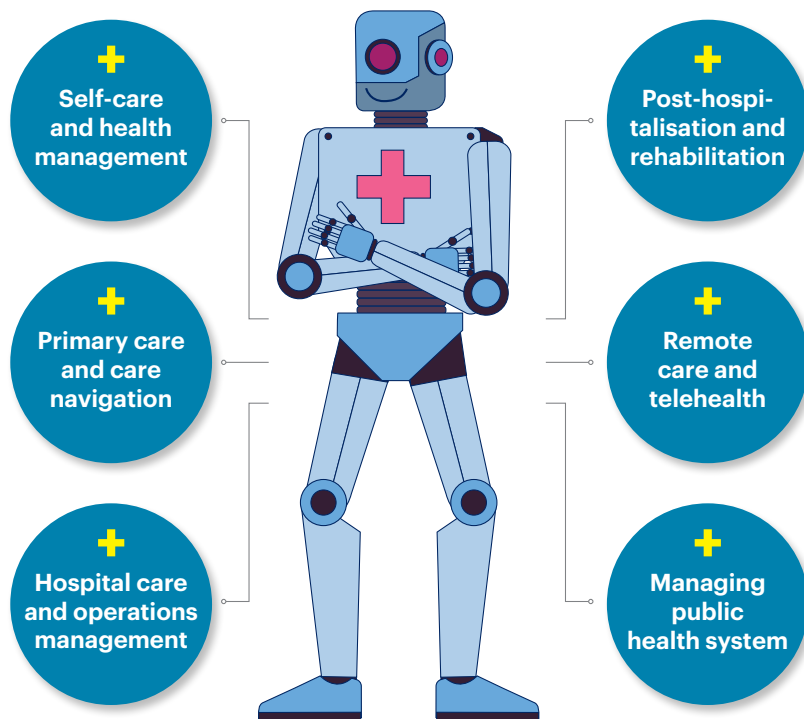
Whether it is radiology, pathology, cardiology or neurology, these specialties generate vast volumes of structured data, making them ideal for AI-driven analysis, he says. More importantly, AI has moved beyond being a buzzword in diagnostics as it is increasingly becoming an integral part of everyday clinical workflows.

Will AI replace the medical technician? "The work is highly specialised, data-intensive and constrained by a shortage of experts. In such settings, AI is not replacing clinicians but augmenting them—helping specialists analyse information faster, prioritise cases and improve diagnostic efficiency. That is why diagnostics has emerged as one of the first areas where AI is seeing meaningful, large-scale adoption," says Sivasailam.

Healthcare leaders caution against assuming AI is a solution to every problem. "There are certain sweet spots and low-hanging fruits which can be easily handled by AI," says Vedam Ramprasad, CEO of medical testing lab MedGenome. "But it will not be able to impact every aspect of healthcare."

That distinction is increasingly shaping investment decisions.

EMERGING AI USE CASES



Rather than deploying AI across the board, hospitals and health-tech companies are prioritising areas where structured data, specialist shortages and measurable outcomes create a clear business case.

According to a KPMG report, due to the inherently human-centric nature of certain healthcare functions, AI's impact has been more evolutionary than revolutionary.

In areas such as administrative functions, while AI has the potential to automate several back-office processes, adoption has been slower because of lower labour costs in India.

"For many hospitals, hiring additional staff is more economical than investing in AI-powered

automation, particularly when the cost of deploying advanced AI models remains high. As a result, healthcare providers continue to rely on manpower to solve administrative challenges rather than redesign workflows around AI," says Sivasailam.

MEASURING ROI

Does the use of AI pass the return on investment (ROI) test? Hospitals and health-tech companies are discovering that scaling AI requires years of investment in data, digital infrastructure, governance, cybersecurity, workflow integration and clinician adoption.

Above all, AI has to demonstrate measurable returns in day-to-day business. The conversation has